
MODEL SELECTION AND VALIDATION REPORT (fishing_biomass model)

OBSERVATIONS:1739

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (fishing_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.21	0.24
log_sum_fishing	0.07	0.02
kde_fric	0.19	0.03
kde_feve	-0.23	0.02
tric	0.02	0.02
teve	-0.45	0.02
sbt_mean	0.33	0.07
sic_mean	0.00	0.06
sbt_sd	0.05	0.04
sst_sd	0.04	0.03
log_chla_mean	0.06	0.03
depth	-0.01	0.04

Dispersion parameter: 0.72

Spatiotemporal AR1 correlation (rho): 0.91

Matérn range: 417.10

Spatiotemporal marginal AR1 SD: 0.65

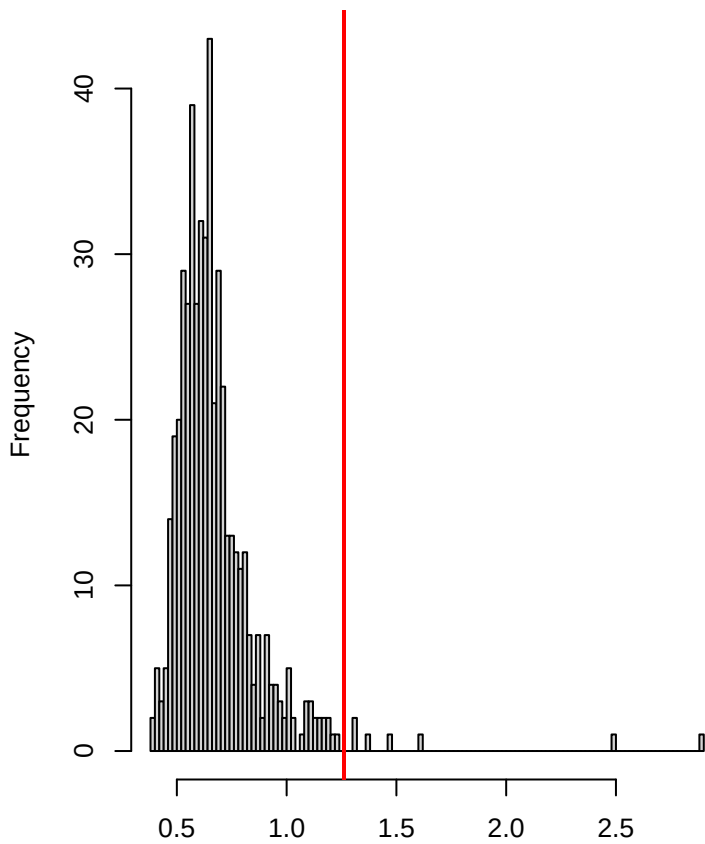
REML criterion at convergence: 14079.326

See ?tidy.sdmTMB to extract these values as a data frame.

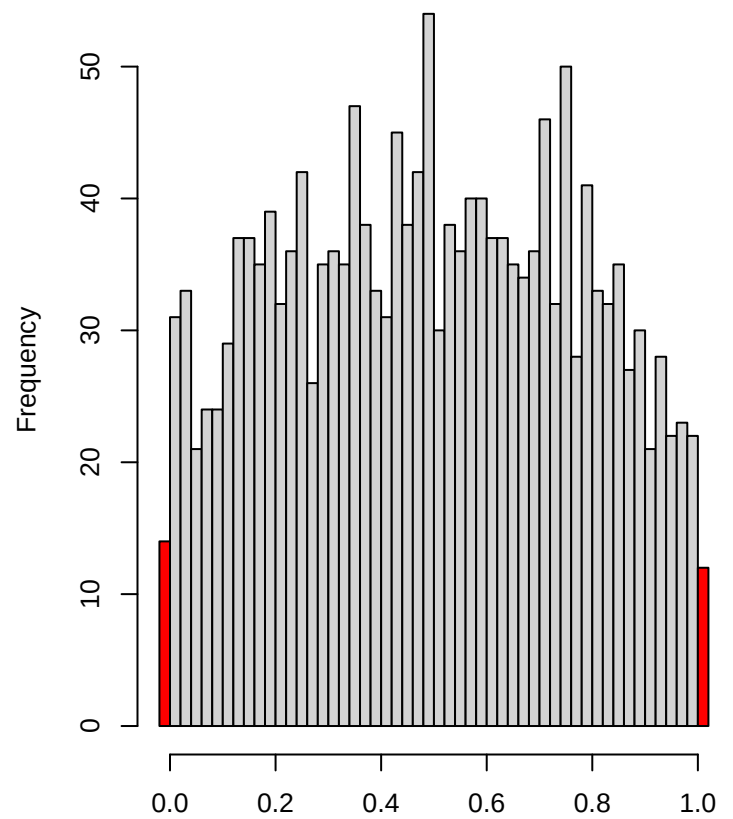
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	16	28190.65	0.3179869
spatiotemporal_random_fields_ar1_with_anisotropy	18	28190.97	89.2376032
spatiotemporal_random_fields_iid_with_anisotropy	17	28280.21	5.3000892
spatiotemporal_random_fields_iid	15	28285.51	21.0102436
spatial_random_fields	15	28306.52	3.0217306
spatial_random_fields_with_anisotropy	17	28309.54	190.5376696
error_only	13	28500.08	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



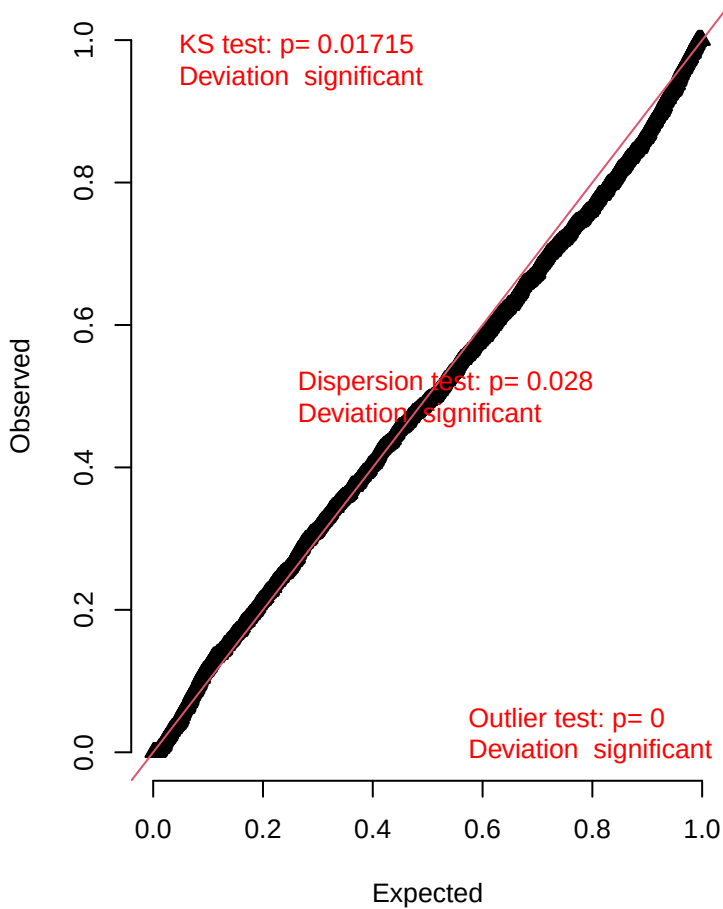
Outlier test significant



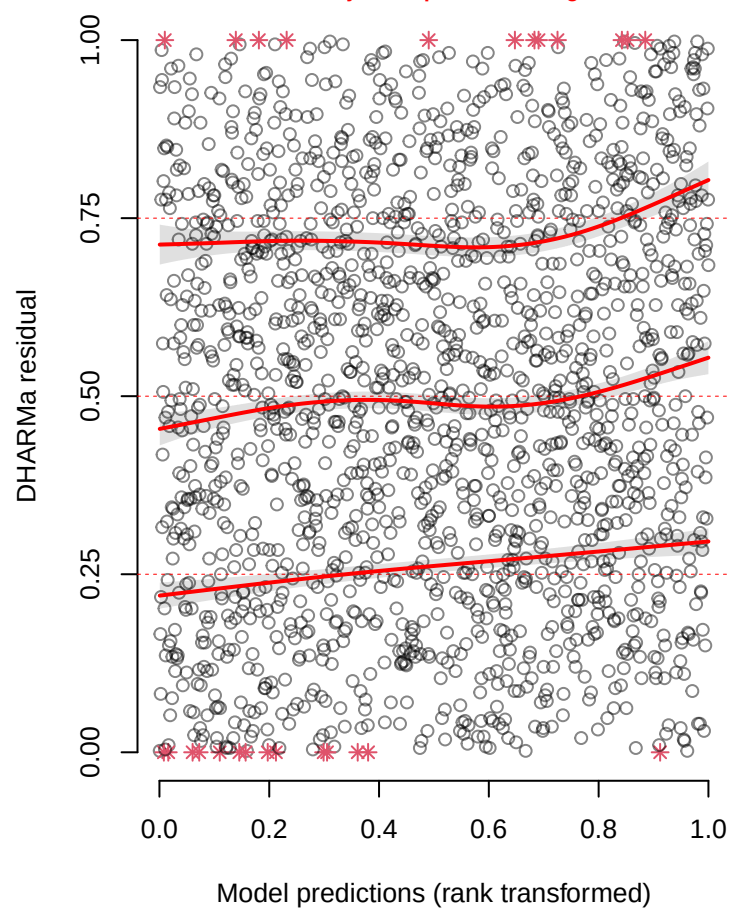
Simulated values, red line = fitted model. p-value (two.sided) =

Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation

Moran's I = 0.015 | H1: Distance-based autocorrelation | p-value = 0.082

